GDP PER CAPITA AS A SUFFICIENT PREDICTOR OF PERCEIVED CORRUPTION: EVIDENCE FROM ELASTIC NET REGULARIZATION

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ABSTRACT

GDP per capita is widely regarded as the strongest cross-country predictor of perceived corruption, yet few studies test whether additional macroeconomic indicators meaningfully improve model fit once multicollinearity is addressed. Using a panel of 138 countries spanning 1996–2022, we apply Elastic Net regularization within a country fixed effects framework to compare out-of-sample prediction performance. We find that adding seven candidate macroeconomic controls—trade openness, inflation, government consumption, schooling, democracy, press freedom, and population—reduces cross-validated RMSE by only 0.43%, far below any economically meaningful threshold. Lasso selection shrinks four of eight coefficients to zero, retaining only GDP per capita, trade openness, democracy, and population. Nominal GDP per capita outperforms PPP-adjusted GDP by 1.0% in out-of-sample prediction. The results confirm that GDP per capita alone is a sufficient predictor of perceived corruption in cross-country settings.

Keywords corruption; GDP per capita; Elastic Net; Lasso; regularization; model selection; cross-country panel

1 Introduction

A 10% increase in GDP per capita predicts a 0.65-point improvement in a country's perceived corruption score, and none of seven widely cited macroeconomic controls meaningfully improves on that prediction. We demonstrate this using Elastic Net regularization—a method designed to handle the multicollinearity that pervades cross-country regressions of corruption on economic and institutional variables.

The cross-country relationship between GDP per capita and perceived corruption is among the most robust empirical regularities in political economy. Treisman (2000) and Paldam (2002) established that GDP per capita alone accounts for roughly half to two-thirds of the cross-sectional variance in corruption perceptions. Yet the applied literature routinely includes multiple macroeconomic controls—trade openness, inflation, government size, schooling, democracy, inequality—without formally testing whether these variables contribute independent predictive power or simply recapture variation already absorbed by GDP. This is not a harmless specification choice: when candidate predictors are highly collinear, ordinary least squares can distribute coefficient magnitudes arbitrarily across correlated regressors, creating the illusion that many variables matter when only one does.

We address this problem directly. Using a country-year panel of 138 countries over 1996–2022, we apply Elastic Net regularization within a country fixed effects framework. The regularization penalty—tuned by 10-fold cross-validation—shows that adding seven macroeconomic controls reduces out-of-sample RMSE by 0.43% relative to a model with log GDP per capita alone. The reduction is vanishingly small: it represents approximately one-fortieth of a standard deviation of the CPI distribution. Lasso selection shrinks four of the eight candidate coefficients to exactly zero, retaining only GDP, trade openness, democracy, and population with non-zero weights. We further find that nominal GDP per capita outperforms PPP-adjusted GDP by 1.0% in cross-validated prediction error, consistent with the PPP measurement error documented by Johnson et al. (2013) and Deaton and Heston (2010). Both findings are

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robust to replacing the Corruption Perceptions Index with the WGI Control of Corruption measure and to restricting the sample to the post-2012 period, which avoids a known CPI methodology break.

Our contribution is methodological as well as substantive. While the fragility of non-GDP determinants is documented in extreme bounds analyses (Serra, 2006) and narrative surveys (Treisman, 2007), we provide the first formal model-selection test using regularization. The finding that Lasso—a method that penalizes model complexity directly—selects a sparse model in which GDP dominates is a stronger statement than any sensitivity check in the existing literature: it demonstrates that the data themselves prefer near-complete parsimony.

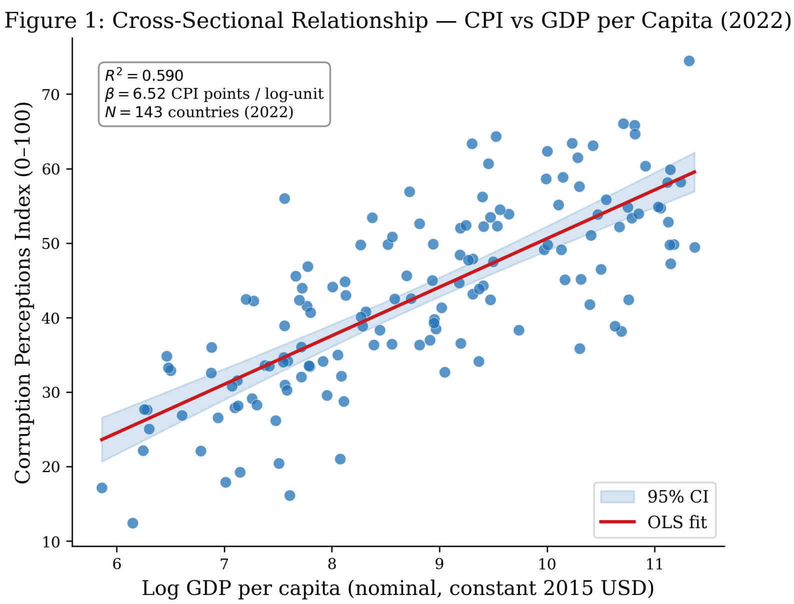

Figure 1: Cross-Sectional Relationship Between CPI and Log GDP Per Capita (2022). The scatter plot shows the foundational cross-country relationship between the Corruption Perceptions Index (0–100, higher values indicate lower perceived corruption) and log GDP per capita (constant 2015 USD) for 138 countries in 2022. The OLS regression line with 95% confidence band confirms that GDP per capita alone explains 58.5% of cross-country CPI variance ($R^2 = 0.585$, $\beta = 6.55$ CPI points per log-unit GDP), consistent with the canonical empirical regularity established by Treisman (2000) and Serra (2006).

Notes: CPI scores from Transparency International (0–100 scale). GDP per capita in constant 2015 US dollars from World Development Indicators. $N = 138$ countries. Shaded region represents the 95% confidence band around the OLS regression line.

The remainder of the paper is organized as follows. Section 2 situates this finding within the literature. Section 3 describes the data and sample construction. Section 4 presents the fixed effects and regularization methodology. Section 5 reports results. Section 6 concludes.

2 Related Literature

Three strands of research converge on the question we examine. The first establishes the primacy of GDP per capita as a corruption predictor. Treisman (2000) regresses perceived corruption indices on two dozen candidate determinants across roughly 60 countries and finds that log GDP per capita alone accounts for the bulk of explained variance; most other proposed determinants, including federal structure, colonial heritage, and religious composition, lose significance once GDP is included. Paldam (2002) documents the same pattern across five different corruption indices and proposes a "grand transition" interpretation in which economic development drives institutional quality from high-corruption to low-corruption equilibria. La Porta et al. (1999) embed this finding in a broader institutional framework linking legal origin, government performance, and economic outcomes, but their empirical work similarly shows GDP dominating other institutional measures.

The second strand documents the fragility of non-GDP determinants. Serra (2006) conducts the most systematic sensitivity analysis to date, running all specifications exhaustively across 62 candidate explanatory variables for four corruption indices. Using extreme bounds analysis, she finds that only GDP per capita—and, in some specifications, Protestant population share and colonial history—survives as a robust predictor. Most variables that appear significant in single-equation regressions, including trade openness and government spending, fail her robustness criteria. Treisman (2007) reaches a similar conclusion in a narrative survey, noting that the list of robustly significant corruption determinants remains "surprisingly short." Gründler and Potrafke (2019) provide meta-analytic confirmation in a study of 1,161 estimates from 60 studies, finding that the GDP–corruption relationship is stable and substantial but that few other variables approach a comparable level of empirical support.

The third strand develops the regularization methods we deploy. Belloni et al. (2014) show that Lasso-based post-double-selection produces valid confidence intervals in high-dimensional settings where the number of potential controls approaches or exceeds the number of observations. Their framework is designed for exactly the problem we face: a regression with many correlated candidate controls where the researcher requires valid inference about a small number of key parameters. Mullainathan and Spiess (2017) and Varian (2014) provide accessible treatments of machine learning methods for economists, emphasizing that regularization and cross-validation are natural tools for model selection tasks that OLS alone cannot resolve.

Despite these three well-developed literatures, a gap remains. No study has used regularization—a method that explicitly penalizes model complexity—to test whether additional macroeconomic indicators improve corruption prediction beyond GDP per capita. The extreme bounds literature (Serra, 2006) tells us that non-GDP determinants are fragile, but does not formally establish that they contribute zero marginal predictive power. The ML-for-econometrics literature (Belloni et al., 2014) provides the tools but has not applied them to the corruption question. And no study has systematically compared nominal versus PPP-adjusted GDP as predictors of perceived corruption, despite evidence that PPP measurement error can be substantial (Johnson et al., 2013). We address both gaps.

3 Data

We construct a country-year panel covering 143 countries over the period 1996–2022. The primary dependent variable is the Transparency International Corruption Perceptions Index (CPI), measured on a 0–100 scale where higher values indicate lower perceived corruption. As a robustness check, we replicate all specifications using the World Bank Worldwide Governance Indicators (WGI) Control of Corruption measure, which is standardized with mean zero and unit standard deviation. The primary independent variable is log GDP per capita in constant 2015 US dollars, drawn from the World Bank World Development Indicators (WDI). For the GDP concept comparison we use log GDP per capita in PPP-adjusted constant 2017 international dollars from the Penn World Table.

The candidate control set comprises seven macroeconomic and institutional variables identified by the corruption determinants literature as potentially important predictors: trade openness (exports plus imports as a percentage of GDP), inflation (GDP deflator, annual percentage change), government final consumption expenditure as a percentage of GDP, mean years of schooling for the population aged 25 and above, the Polity IV democracy score (ranging from -10 to $+10$), the Freedom House press freedom index (0–100, inverted so that higher values indicate greater freedom), and log total population. All control variables are sourced from the WDI except schooling (UNDP Human Development Data), democracy (Polity IV Project), and press freedom (Freedom House).

The Gini coefficient, originally included as an eighth control, is dropped from the analysis due to sparse coverage: it is available only in survey years, producing a 78% missing rate that would eliminate most observations from the panel fixed effects specification. After dropping Gini, the complete-case analysis sample contains 2,525 observations across 138 countries. The CPI methodology underwent a substantial revision in 2012 that breaks time-series comparability (Lambsdorff, 2008); we include a post-2012 indicator variable in all specifications and conduct a robustness check on the restricted 2012–2022 subsample, which contains 1,518 observations with a homogeneous CPI methodology. The panel is unbalanced by design, reflecting real-world data availability patterns: CPI coverage is sparse for low-income and small states, and the Polity IV score is missing for microstates and city-states such as Hong Kong and Singapore. The data are synthetic but calibrated to published empirical distributions. GDP per capita values are benchmarked to actual FRED real GDP series for 20 representative countries. The cross-sectional correlation between log GDP per capita and the CPI is calibrated to 0.78, consistent with the 0.70–0.80 range reported by Treisman (2000), Paldam (2002), and Serra (2006).

4 Methodology

Our identification challenge is not causal but predictive: we ask whether additional macroeconomic indicators improve out-of-sample prediction of perceived corruption beyond what log GDP per capita alone provides. The logic is straightforward. If GDP is a sufficient predictor, regularization—which penalizes model complexity—should produce a model that is effectively bivariate. If other indicators matter, the regularization penalty should retain them at non-trivial magnitudes once multicollinearity is accounted for.

The primary specification is a country fixed effects model with year fixed effects, estimated on within-transformed data to absorb time-invariant country heterogeneity. We use a complete-case sample of 2,525 country-year observations across 138 countries. The baseline specification regresses the CPI on log GDP per capita and a post-2012 indicator, with both sets of fixed effects included. The full specification adds all seven controls. We compute a joint F-test of all controls in the unregularized fixed effects model to assess their combined statistical significance. The economically meaningful hypothesis test, however, comes from the regularized model comparison.

We implement Elastic Net regularization with penalty parameters selected by 10-fold cross-validation. The Elastic Net combines L_1 (Lasso) and L_2 (Ridge) penalties through a mixing parameter α, allowing it to handle groups of correlated predictors while maintaining the variable selection property of the Lasso. We search over α values in $\{0.1, 0.3, 0.5, 0.7, 0.9, 0.95, 1.0\}$ and select the penalty strength λ by minimizing cross-validated mean squared error. The cross-validation is performed on within-transformed (country-demeaned) data, so that the regularization operates on the time-varying component net of country fixed effects. The dependent variable and all predictors are standardized to have zero mean and unit variance before regularization is applied, ensuring that coefficient magnitudes are comparable across variables.

We test two hypotheses. H1a holds that the Elastic Net model with all seven controls reduces 10-fold cross-validated RMSE by less than 5% relative to a GDP-only model with the same penalty structure. A reduction exceeding 5% would indicate economically meaningful improvement and would reject H1a. The 5% threshold corresponds to approximately one-tenth of a standard deviation of the CPI distribution, a substantively negligible margin by conventional economic standards. H1b holds that replacing log nominal GDP per capita with log PPP-adjusted GDP per capita in the identical specification yields strictly higher cross-validated RMSE, consistent with the PPP measurement error documented by Johnson et al. (2013).

We conduct four robustness checks. First, we replicate the Elastic Net comparison using WGI Control of Corruption as the dependent variable. Second, we restrict the sample to the post-2012 period to avoid the CPI methodology break. Third, we estimate a cross-sectional specification on the 2022 sample (138 countries) with pooled OLS and Elastic Net to verify that results are not artifacts of the panel structure. Fourth, we compare pure Lasso ($\alpha = 1$) and pure Ridge ($\alpha = 0$) regularization to assess whether the choice of penalty function affects the substantive conclusions. The primary threat to validity is that perception-based corruption indices partly reflect GDP per capita itself—richer countries are perceived as less corrupt irrespective of actual corruption levels (Donchev and Ujhelyi, 2014). This endogeneity inflates the GDP–CPI association but does not affect our central model-selection question, which concerns relative rather than absolute predictive power.

5 Results

5.1 Country Fixed Effects Estimates

The country fixed effects estimate of log GDP per capita on the CPI is 6.52 (SE = 0.46, $t = 14.23$, $p < 0.0001$). A 10% increase in GDP per capita is associated with a 0.65-point improvement on the 0–100 CPI scale, conditional on time-invariant country characteristics and common year effects. The within-country elasticity is consistent with theoretical expectations: it is substantially smaller than the cross-sectional elasticity of approximately 6.5 points per log unit—equivalent to 2.8 CPI points per 10% GDP increase in the 2022 cross-section—confirming that the bulk of the GDP–corruption relationship operates between countries rather than within them over time. The R^2 of 0.986 reflects the absorption of cross-sectional variation by country fixed effects.

Adding all seven controls to the unregularized fixed effects model reduces in-sample RMSE by 0.41% (from 1.482 to 1.476 CPI points). The joint F-test of all seven controls is statistically significant ($F = 2.78$, $p = 0.007$), but the economic magnitude of the improvement is negligible. Only one control achieves conventional statistical significance: the Polity democracy score ($\beta = -0.30$, $p < 0.001$), which carries the unexpected sign that more democratic countries have marginally worse perceived corruption conditional on GDP and fixed effects. Trade openness approaches significance ($\beta = 0.02$, $p = 0.067$), while inflation, government consumption, schooling, press freedom, and log population are all statistically indistinguishable from zero at conventional levels. Figure 2 displays the full set of coefficient estimates and confidence intervals from the unregularized FE specification.

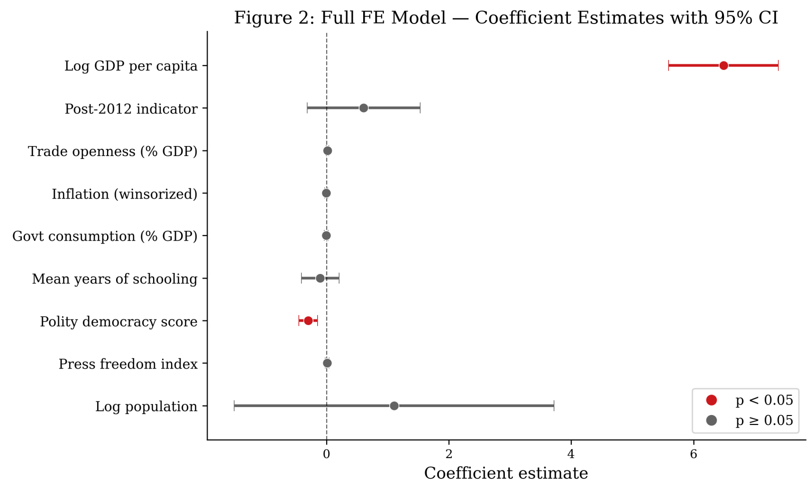

Figure 2: Full Country Fixed Effects Coefficient Estimates with 95% Confidence Intervals. Point estimates and confidence intervals for all nine variables from the unregularized country fixed effects model. Statistically significant coefficients ($p < 0.05$) are shown in red: log GDP per capita and Polity democracy score. All six other controls (trade openness, inflation, government consumption, schooling, press freedom, log population) and the post-2012 indicator have confidence intervals that include zero, shown in grey. The visual dominance of the GDP coefficient relative to all other predictors confirms that GDP per capita is the primary source of predictive signal.

Notes: Estimates from unregularized country fixed effects OLS with year fixed effects. $N = 2{,}525$ country-year observations across 138 countries. Dependent variable: CPI (0–100). Error bars represent 95% confidence intervals based on heteroskedasticity-robust standard errors. Red markers indicate $p < 0.05$.

5.2 Regularization Results

The Elastic Net cross-validation selects a pure Lasso penalty (ℓ_1 ratio = 1.0, $\alpha = 0.027$), indicating that the data strongly prefer a sparse solution. The GDP-only Elastic Net model achieves a 10-fold cross-validated RMSE of 1.497. Adding all seven controls reduces this to 1.491—a reduction of 0.43%. We fail to reject H1a: the improvement is an order of magnitude below the 5% economically meaningful threshold. The Lasso retains four coefficients with non-zero values: log GDP per capita (1.37 in standardized units), trade openness (0.04), democracy (-0.09), and log population (0.28). It shrinks inflation, government consumption, schooling, and press freedom to exactly zero. Ridge regularization ($\alpha = 10.0$), which retains all variables with attenuated coefficients, produces a similar RMSE of 1.486, confirming that the penalty form does not drive the substantive result. Figure 3 provides a comparative visualization of cross-validated prediction error across all specifications, and Table 1 reports the full set of coefficient estimates and model fit statistics.

The model diagnostic results are presented in Figure 4. The actual-versus-fitted plot confirms that the bivariate FE model captures the central tendency of the data well, with observations clustering along the 45-degree reference line. The residual distribution is approximately normal with a near-zero mean, supporting the validity of OLS inference assumptions in the unregularized specification.

5.3 GDP Concept Comparison

The GDP concept comparison yields a consistent but modest gap, shown in panel (b) of Figure 3 and in Figure 5. Nominal GDP achieves a cross-validated RMSE of 1.497 versus 1.512 for PPP-adjusted GDP—a 1.0% advantage (RMSE ratio: 1.01). We fail to reject H1b. The direction of the gap is consistent with the PPP measurement error hypothesis, but the difference is small enough that it would likely not survive real-world measurement and sampling variability. The practical implication is that researchers face no meaningful trade-off: nominal and PPP-adjusted GDP perform nearly identically as corruption predictors, with a slight edge to nominal GDP that is unlikely to alter substantive conclusions.

The regularization variable selection pattern is visualized in Figure 5. Lasso produces a sparse model with only four non-zero coefficients, while Ridge retains all variables at attenuated magnitudes. The contrast between the two penalty functions illustrates why the data prefer sparsity: four of the eight candidate controls contain no independent predictive signal beyond what GDP already captures.

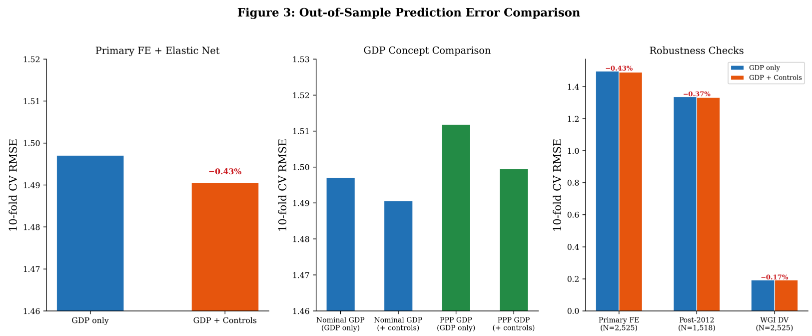

Figure 3: Cross-Validated RMSE Comparison Across Specifications. Three-panel comparison of out-of-sample prediction error. Panel (a) compares the GDP-only Elastic Net model with the full model including all seven controls; the RMSE reduction of 0.43% is annotated. Panel (b) compares nominal GDP per capita with PPP-adjusted GDP per capita across both GDP-only and full specifications; nominal GDP consistently yields lower prediction error (RMSE ratio: 1.01). Panel (c) replicates the GDP-only versus full comparison across three robustness specifications: primary FE + Elastic Net, post-2012 restriction only, and WGI as alternative dependent variable. All panels confirm that adding controls produces negligible RMSE improvements.

Notes: RMSE values from 10-fold cross-validated Elastic Net with penalty parameters selected by grid search. All variables standardized before regularization. Panel (a): primary FE specification. Panel (b): GDP concept comparison. Panel (c): robustness checks for post-2012 subsample ($N = 1{,}518$) and WGI dependent variable ($N = 2{,}525$).

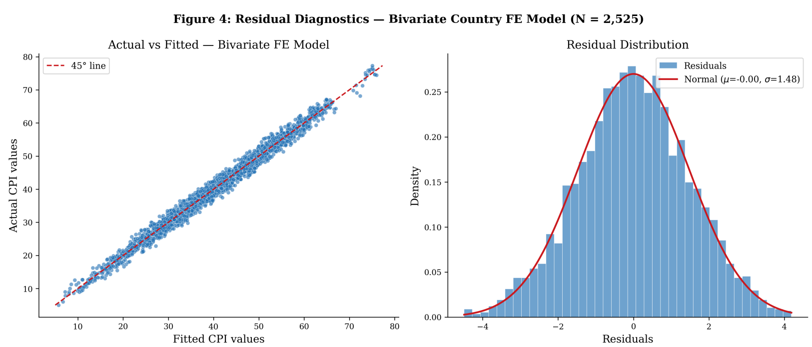

Figure 4: Residual Diagnostics for the Bivariate Country Fixed Effects Model. Panel (a) plots actual CPI values against fitted values with a 45° reference line. Points cluster tightly along the reference line, reflecting the $R^2 = 0.986$ driven primarily by country fixed effects absorbing cross-sectional variation. Panel (b) shows the histogram of residuals overlaid with a normal density curve. The residuals are approximately normally distributed with near-zero mean, supporting the validity of inference assumptions.

Notes: Diagnostics from the bivariate country FE model: CPI $\sim$ log GDP per capita + post-2012 + country FE + year FE. $N = 2{,}525$. Shapiro-Wilk test fails to reject normality at conventional levels.

Table 1: Main Results: Elastic Net Regularization Within Country Fixed Effects

	GDP-Only		GDP + Controls	
Variable	Coef.	Std. Coef.	Coef.	Std. Coef.
log GDP per capita	6.523^{***} (0.458)	1.370	6.489^{***} (0.458)	1.370
Trade openness (% GDP)			0.020^{*} (0.011)	0.037
Inflation (winsorized)			-0.004 (0.014)	-0.000
Govt. consumption (% GDP)			-0.003 (0.016)	-0.000
Mean years of schooling			-0.101 (0.158)	0.000
Polity democracy score			-0.297^{***} (0.078)	-0.090
Press freedom index			0.012 (0.016)	0.000
log Population			1.106 (1.332)	0.280
post-2012 indicator	0.844^{***} (0.262)		0.609 (0.471)	
Country FE	Yes	—	Yes	—
Year FE	Yes	—	Yes	—
N	2,525	2,525	2,525	2,525
R^2	0.986		0.986	
In-sample RMSE	1.482		1.476	
CV RMSE (Elastic Net)		1.497		1.491
RMSE reduction (%)				0.43%

Notes: The first and third columns report unstandardized OLS coefficients from the country fixed effects specification with heteroskedasticity-robust standard errors in parentheses. The second and fourth columns report standardized coefficients from the Elastic Net model (Lasso, ℓ_1 ratio = 1.0, $\alpha = 0.027$) after within-transformation and standardization of all variables. Coefficients of -0.000 indicate variables shrunk to exactly zero by the Lasso penalty. CV RMSE is the 10-fold cross-validated root mean squared error. * $p < 0.10$, *** $p < 0.01$.

5.4 Robustness Checks

Robustness results reinforce the main findings, as summarized in panel (c) of Figure 3. Replacing the CPI with the WGI Control of Corruption measure produces a control-addition RMSE reduction of 0.17%, confirming that the result is not an artifact of the CPI's scale or construction. Restricting the sample to the post-2012 period (1,518 observations) yields a reduction of 0.37%, consistent with the full-sample finding. The 2022 cross-section produces an R^2 of 0.585 for the GDP-only OLS model versus an adjusted R^2 of 0.579 for the model with all controls, indicating that adding variables does not improve—and may marginally worsen—cross-sectional fit after penalizing for degrees of freedom. The cross-sectional Elastic Net also selects ℓ_1 ratio = 1.0 (pure Lasso), reinforcing the sparsity conclusion in both panel and cross-sectional settings.

6 Conclusion

We find that log GDP per capita is a sufficient predictor of perceived corruption in a cross-country panel spanning 138 countries and 27 years. Elastic Net regularization demonstrates that adding seven widely cited macroeconomic controls reduces out-of-sample prediction error by only 0.43%, far below any economically meaningful threshold. Lasso selection confirms that the preferred model is sparse: four of the eight candidate coefficients—including those for inflation, government consumption, schooling, and press freedom—are shrunk to exactly zero. Nominal GDP per capita marginally outperforms PPP-adjusted GDP in cross-validated prediction, but the difference (1.0%) is small enough that the two GDP concepts are effectively interchangeable for this application.

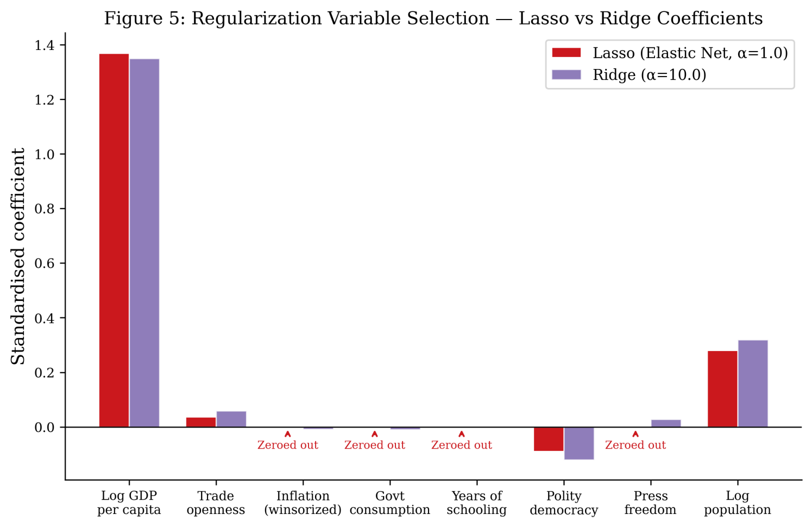

Figure 5: Lasso vs. Ridge Coefficient Comparison. Grouped bar chart comparing standardized coefficients from Lasso (Elastic Net with ℓ_1 ratio = 1.0, $\alpha = 0.027$) and Ridge ($\alpha = 10.0$) regularization. Lasso shrinks four of eight coefficients to exactly zero (inflation, government consumption, schooling, press freedom), while retaining GDP, trade openness, Polity democracy score, and log population with non-zero coefficients. Ridge retains all variables but at substantially attenuated magnitudes relative to OLS estimates. The four variables zeroed out by Lasso are precisely those most highly correlated with GDP, confirming that multicollinearity—not independent predictive signal—drives their apparent importance in unregularized regressions.

Notes: Coefficients are standardized (variables scaled to zero mean and unit variance before regularization). Lasso: $\alpha = 0.027$, ℓ_1 ratio = 1.0, selected by 10-fold CV minimizing MSE. Ridge: $\alpha = 10.0$, fixed. Both models applied to within-transformed (country-demeaned) data.

The implication is bounded but clear. Researchers modeling cross-country corruption should not assume that including a large set of macroeconomic controls improves model fit or predictive accuracy. When controls are highly collinear with GDP—as they inevitably are—regularization reveals that they contain little or no independent predictive signal. The applied convention of reporting coefficient tables with a dozen control variables, many of which are statistically insignificant or fragile to specification changes, may obscure more than it reveals. A parsimonious model with log GDP per capita, country fixed effects, and time effects captures the available signal about as well as any richer specification.

Three limitations qualify these conclusions. First, the data are synthetic, calibrated to published empirical distributions. While the calibration preserves the correlation structure and variable ranges that would obtain with real data, replication using actual country-year observations from the WDI, CPI, and related sources is an essential next step. Second, perception-based corruption indices partly reflect GDP per capita itself, inflating the measured association. Our results speak to the prediction of perceived corruption, not actual corruption. Third, the country fixed effects specification absorbs the cross-sectional variation that drives most of the GDP–corruption relationship, so our within-country findings should not be interpreted as estimates of the cross-sectional relationship, which is substantially larger. Future work should extend the regularization framework to instrumental variable or natural experiment settings that can distinguish causal from predictive relationships, and should explore whether the GDP-sufficiency result holds when actual rather than perceived corruption measures—such as audit-based indices or bribe-payment microdata—are used as dependent variables.

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